

Natural Language Processing course

- This is the 2025 iteration of the course, materials are added as we prepare them.
 - Lecture and seminar materials for each week are in ./week* folders, see README.md for materials and instructions
 - Any technical issues, ideas, bugs in course materials, contribution ideas - add an [issue](#)
- Installing libraries and troubleshooting: [this thread](#).

Syllabus

- **[week01](#) Word Embeddings**
 - Lecture: Word embeddings. Distributional semantics. Count-based (pre-neural) methods. Word2Vec: learn vectors. GloVe: count, then learn. Evaluation: intrinsic vs extrinsic. Analysis and Interpretability. [Interactive lecture materials and more.](#)
 - Seminar: Playing with word and sentence embeddings
 - Homework: Embedding-based machine translation system
- **[week02](#) Language Modeling**
 - Lecture: Language Modeling: what does it mean? Left-to-right framework. N-gram language models. Neural Language Models: General View, Recurrent Models, Convolutional Models. Evaluation. Practical Tips: Weight Tying. Analysis and Interpretability. [Interactive lecture materials](#)

[and more.](#)

- Seminar: Build a N-gram language model from scratch
- Homework: Neural LMs & smoothing in count-based models.
- **[week03](#) Seq2seq and Attention**
 - Lecture: Seq2seq Basics: Encoder-Decoder framework, Training, Simple Models, Inference (e.g., beam search). Attention: general, score functions, models. Transformer: self-attention, masked self-attention, multi-head attention; model architecture. Subword Segmentation (BPE). Analysis and Interpretability: functions of attention heads; probing for linguistic structure. [Interactive lecture materials and more.](#)
 - Seminar: Basic sequence to sequence model
 - Homework: Machine translation with attention
- **[week04](#) Transfer Learning**
 - Lecture: What is Transfer Learning? Great idea 1: From Words to Words-in-Context (CoVe, ELMo). Great idea 2: From Replacing Embeddings to Replacing Models (GPT, BERT). (A Bit of) Adaptors. Analysis and Interpretability. [Interactive lecture materials and more.](#)
 - Homework: fine-tuning a pre-trained BERT model
- **[week05](#) Large Language Models**
 - Lecture: Scaling laws. Emergent abilities. Open-source LLMs.

- Practice: hands-on with open-source LLMs
- **week06 Prompting & In-Context Learning**
 - Lecture: Prompting techniques. Chain-of-Thought reasoning. In-context learning: how and why it works. Analysis and Interpretability.
 - Homework: manual prompt engineering and chain-of-thought reasoning
- **week07 Fine-tuning (PEFT & RLHF)**
 - Lecture: Parameter-efficient fine-tuning (LoRA, adapters). Reinforcement Learning from Human Feedback (RLHF).
 - Seminar + Homework
- **week08 Efficiency**
 - Lecture: Quantization. Distillation. Pruning. Speculative decoding.
 - Homework
- **week09 Retrieval-Augmented Generation (RAG)**
 - Lecture: Dense retrieval. RAG architectures.
 - Practice
- **week10 AI Agents**
 - Lecture: Agent architectures. Tool use. Memory.
 - Seminar + Homework
- **week11 Interpretability**
 - Lecture: Probing. Mechanistic interpretability.
 - Seminar + Homework
- **week12 Multimodal LLMs**
- **week13 Building LLM Systems**
- **week14 AI Agents in Production**